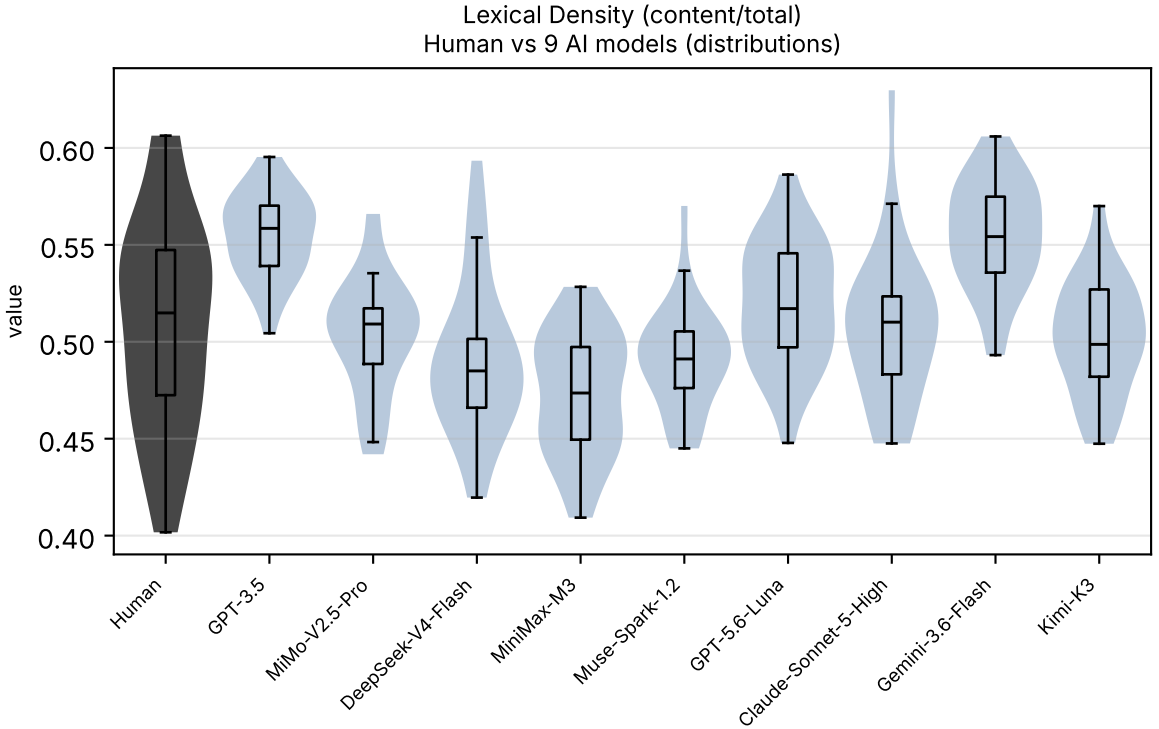
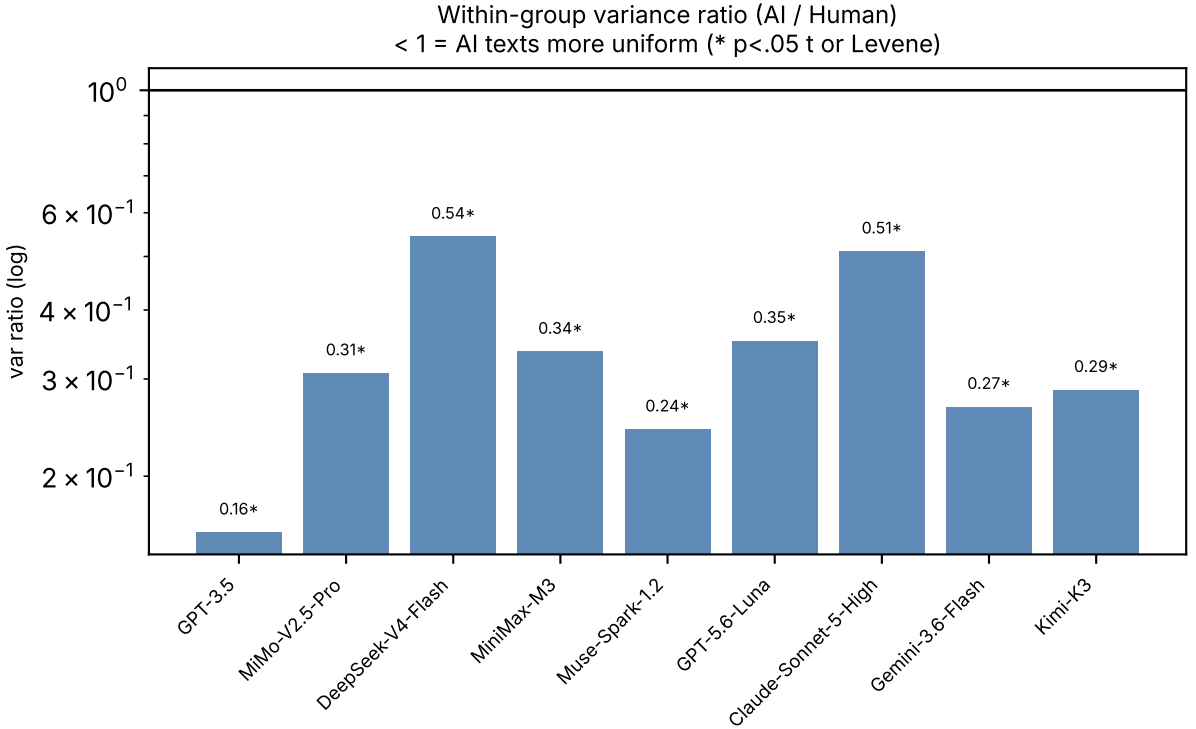
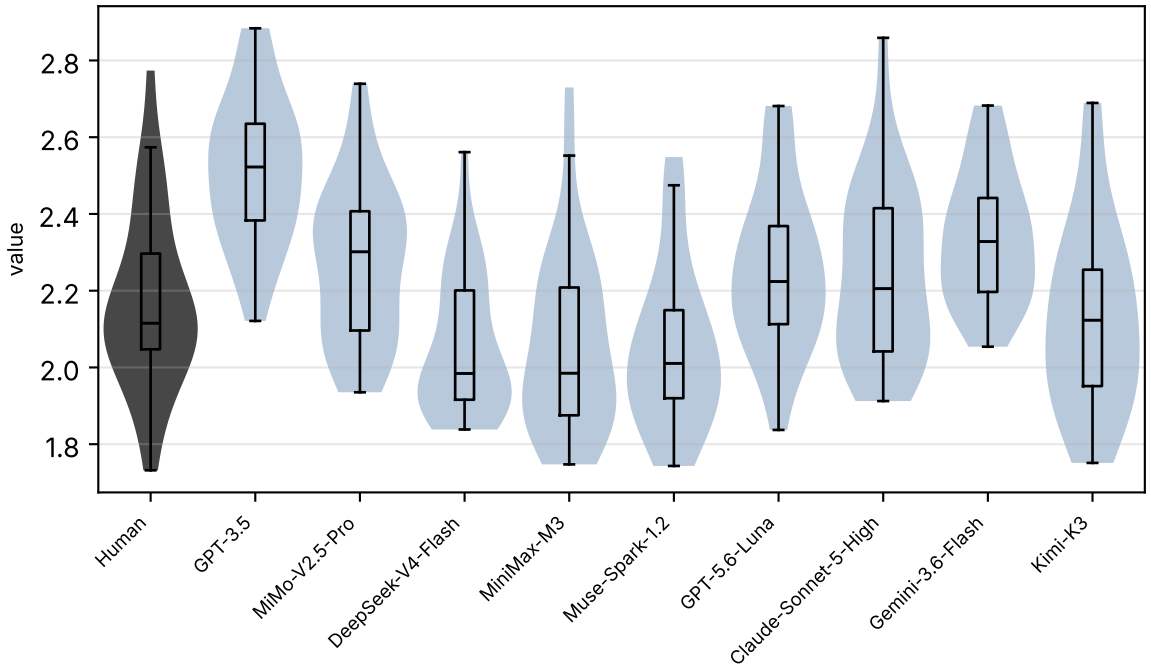
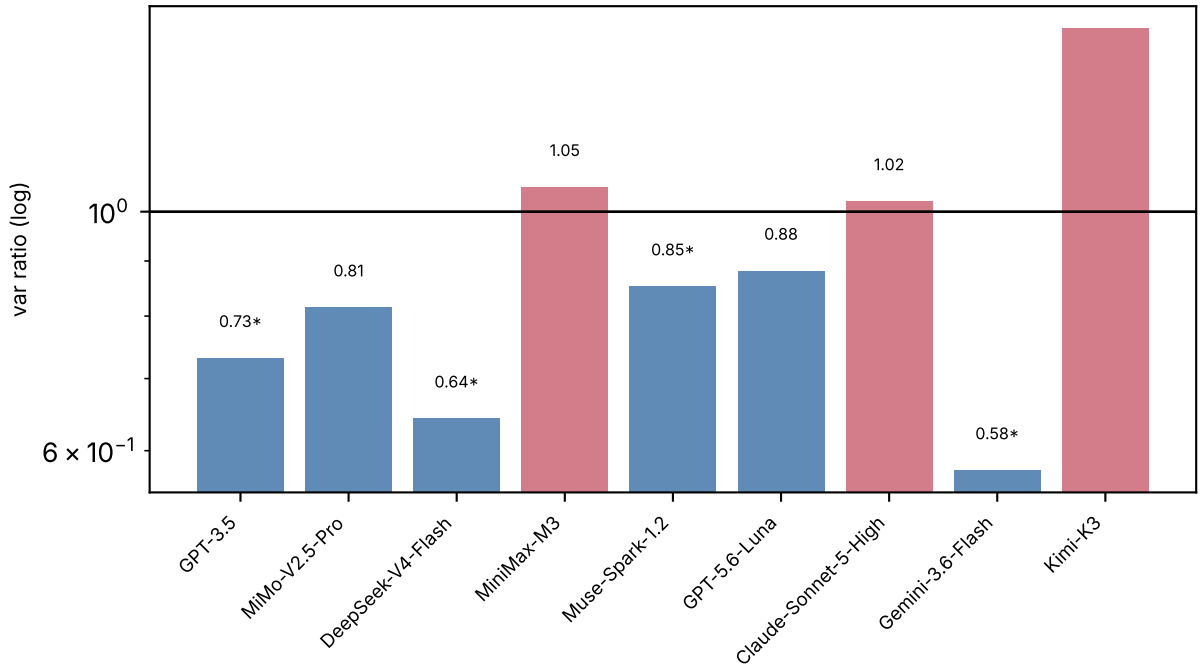
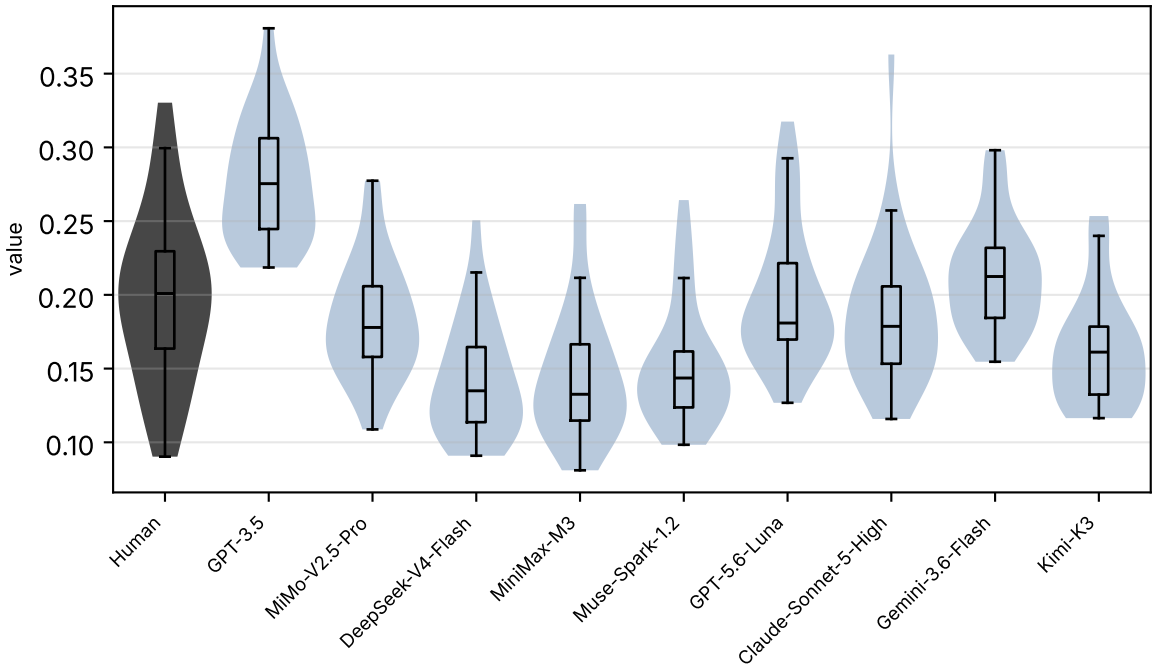
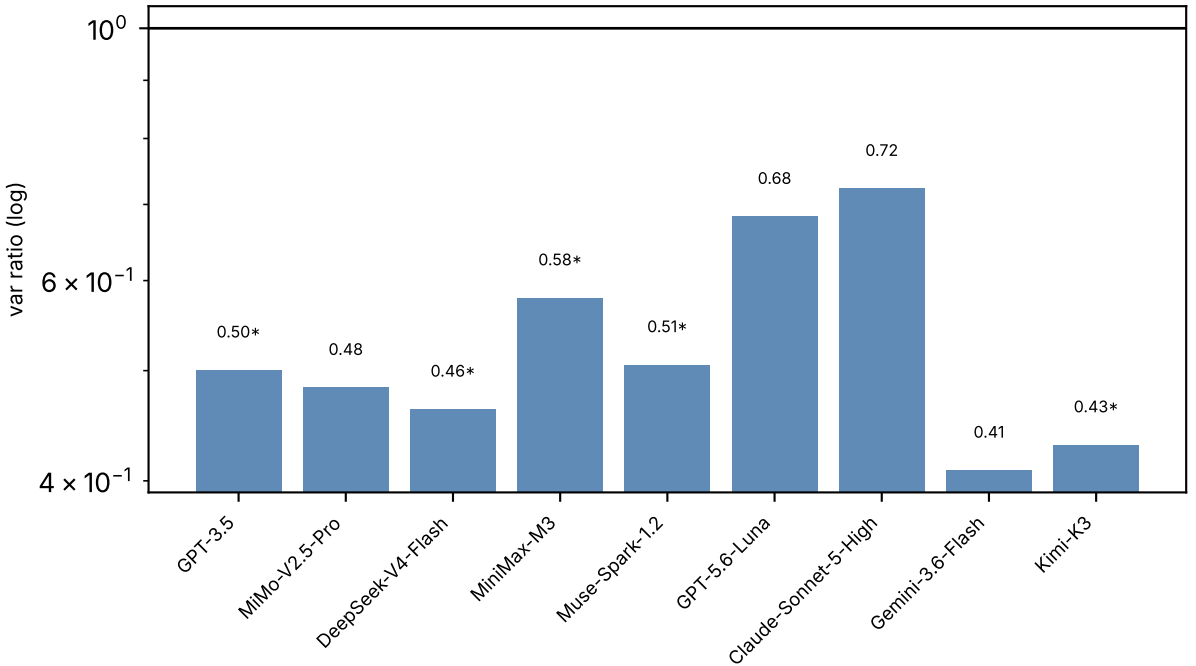

Lexical Density (content/total)
Human vs 9 AI models (distributions)
value
0.60
0.55
0.50
0.45
0.40
Human
GPT-3.5
MiMo-V2.5-Pro
DeepSeek-V4-Flash
MiniMax-M3
Muse-Spark-1.2
GPT-5.6-Luna
Claude-Sonnet-5-High
Gemini-3.6-Flash
Kimi-K3

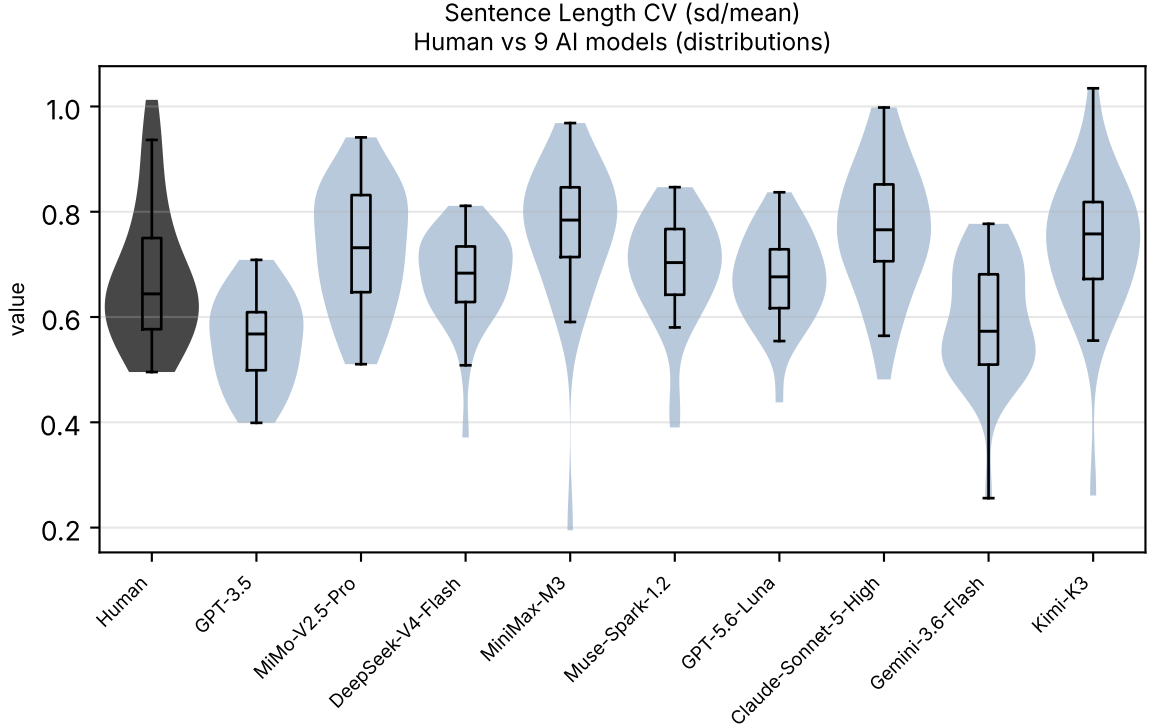

Within-group variance ratio (AI / Human)
< 1 = AI texts more uniform (* p<.05 t or Levene)
var ratio (log)
10^{0}
6×10^{-1}
4×10^{-1}
3×10^{-1}
2×10^{-1}
0.16*
0.31*
0.54*
0.34*
0.24*
0.35*
0.51*
0.27*
0.29*
GPT-3.5
MiMo-V2.5-Pro
DeepSeek-V4-Flash
MiniMax-M3
Muse-Spark-1.2
GPT-5.6-Luna
Claude-Sonnet-5-High
Gemini-3.6-Flash
Kimi-K3

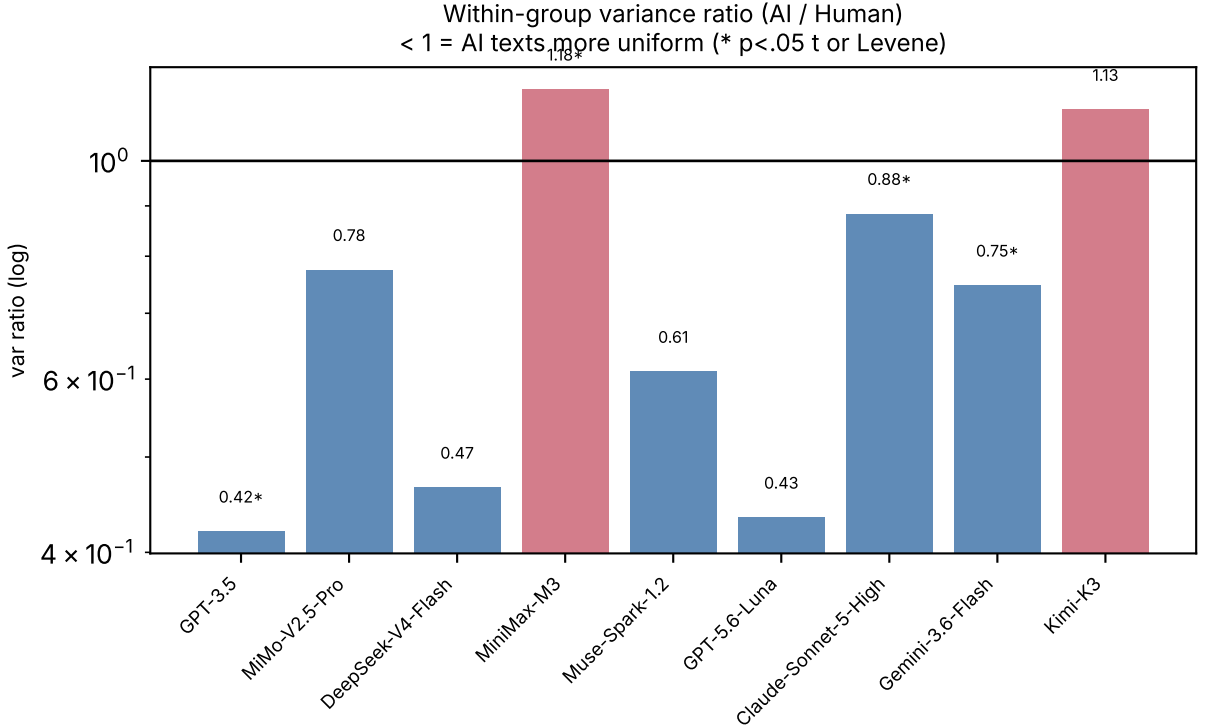

Word Length SD
Human vs 9 AI models (distributions)
value
2.8
2.6
2.4
2.2
2.0
1.8
Human
GPT-3.5
MiMo-V2.5-Pro
DeepSeek-V4-Flash
MiniMax-M3
Muse-Spark-1.2
GPT-5.6-Luna
Claude-Sonnet-5-High
Gemini-3.6-Flash
Kimi-K3

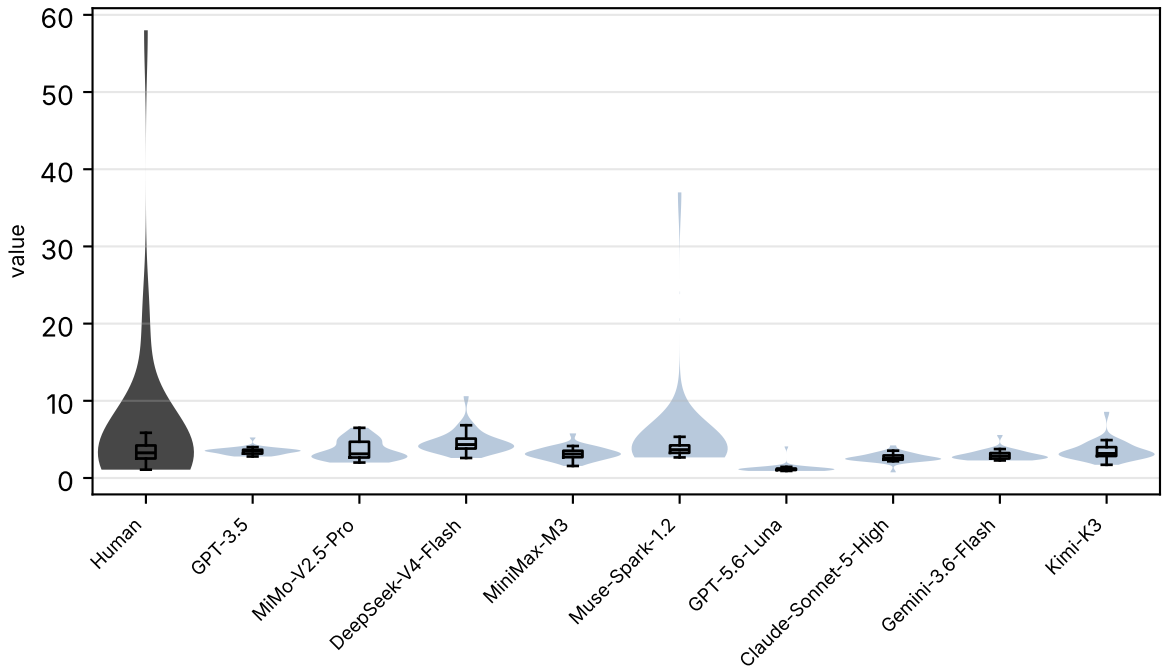

Within-group variance ratio (AI / Human)
< 1 = AI texts more uniform (* p<.05 t or Levene)
var ratio (log)
10^0
6×10^{-1}
0.73*
0.81
0.64*
1.05
0.85*
0.88
1.02
0.58*
1.48
GPT-3.5
MiMo-V2.5-Pro
DeepSeek-V4-Flash
MiniMax-M3
Muse-Spark-1.2
GPT-5.6-Luna
Claude-Sonnet-5-High
Gemini-3.6-Flash
Kimi-K3

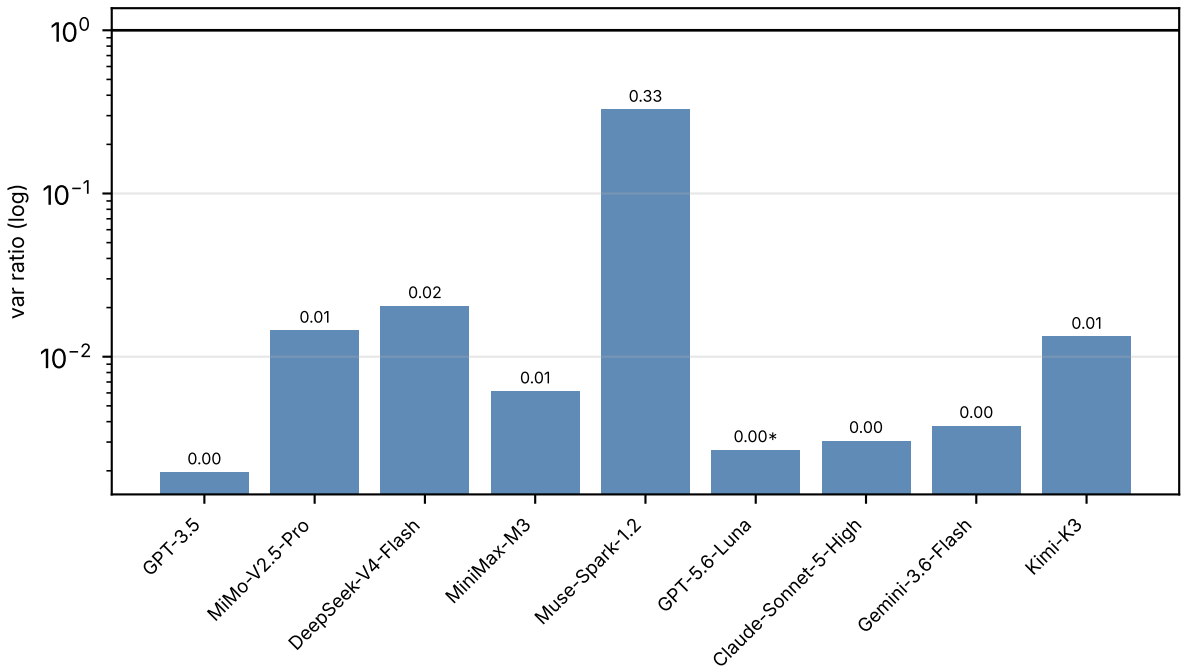

Long Word Ratio (>=7 chars)
Human vs 9 AI models (distributions)
value
0.35
0.30
0.25
0.20
0.15
0.10
Human
GPT-3.5
MiMo-V2.5-Pro
DeepSeek-V4-Flash
MiniMax-M3
Muse-Spark-1.2
GPT-5.6-Luna
Claude-Sonnet-5-High
Gemini-3.6-Flash
Kimi-K3

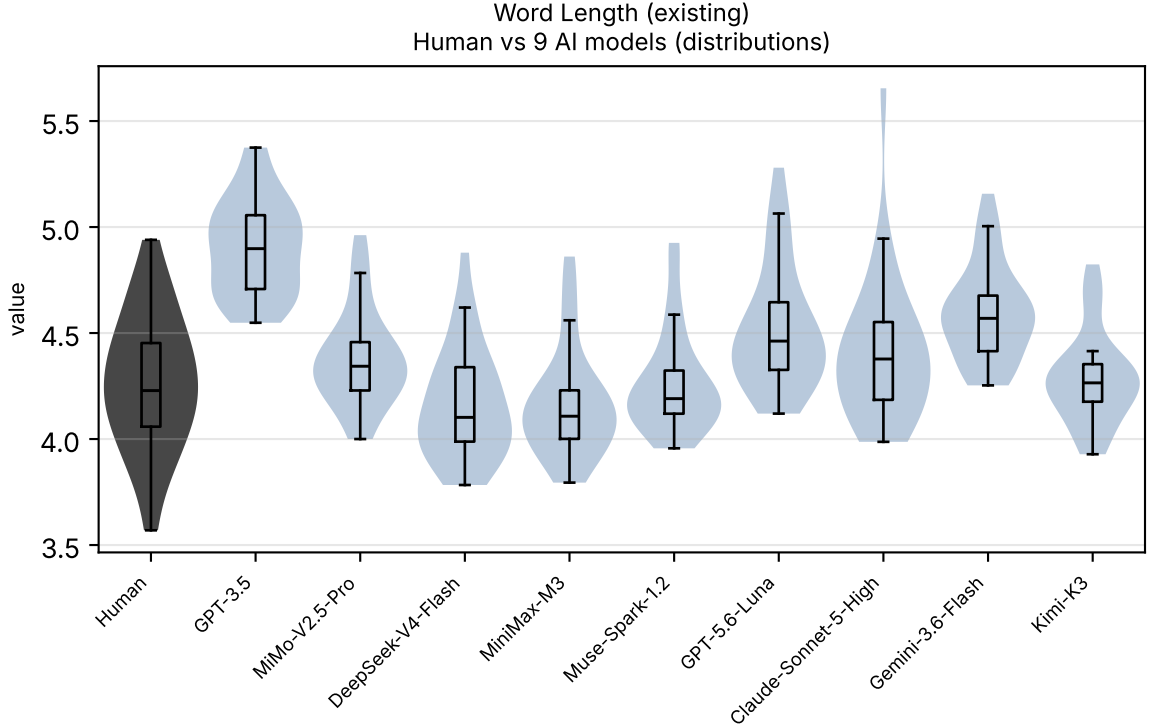

Within-group variance ratio (AI / Human)
< 1 = AI texts more uniform (* p<.05 t or Levene)
var ratio (log)
10^{0}
6×10^{-1}
4×10^{-1}
0.50*
0.48
0.46*
0.58*
0.51*
0.68
0.72
0.41
0.43*
GPT-3.5
MiMo-V2.5-Pro
DeepSeek-V4-Flash
MiniMax-M3
Muse-Spark-1.2
GPT-5.6-Luna
Claude-Sonnet-5-High
Gemini-3.6-Flash
Kimi-K3

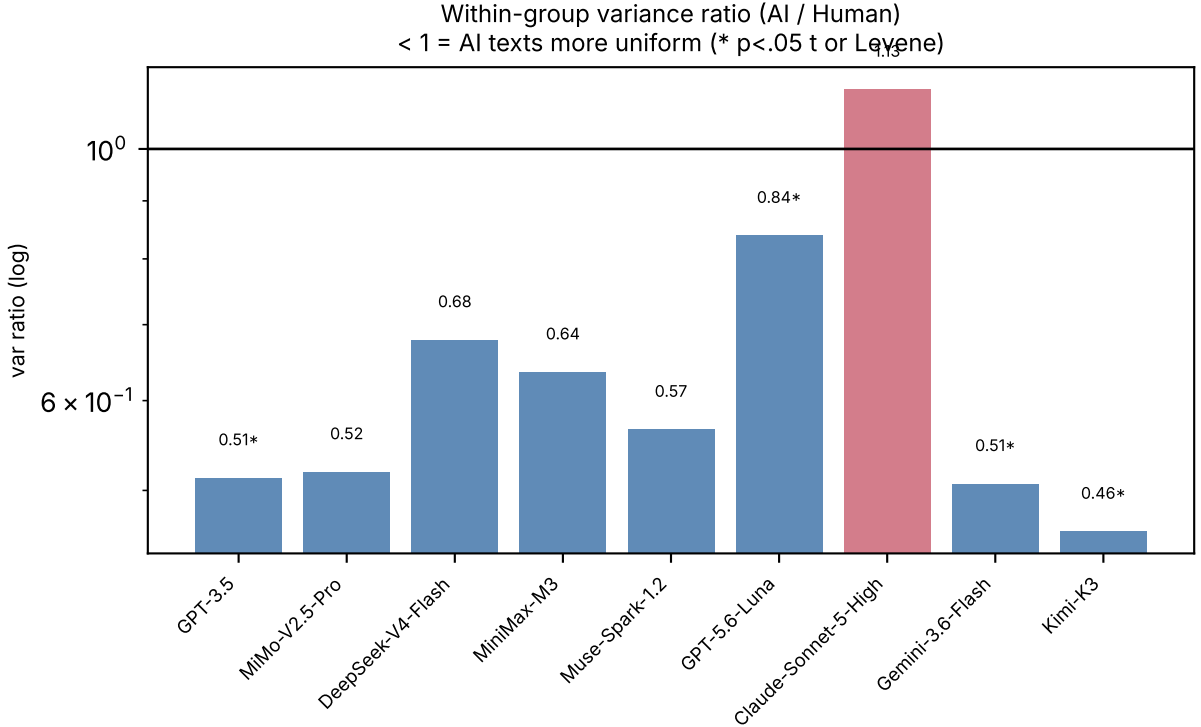

Sentence Length CV (sd/mean)
Human vs 9 AI models (distributions)
value
1.0
0.8
0.6
0.4
0.2
Human
GPT-3.5
MiMo-V2.5-Pro
DeepSeek-V4-Flash
MiniMax-M3
Muse-Spark-1.2
GPT-5.6-Luna
Claude-Sonnet-5-High
Gemini-3.6-Flash
Kimi-K3

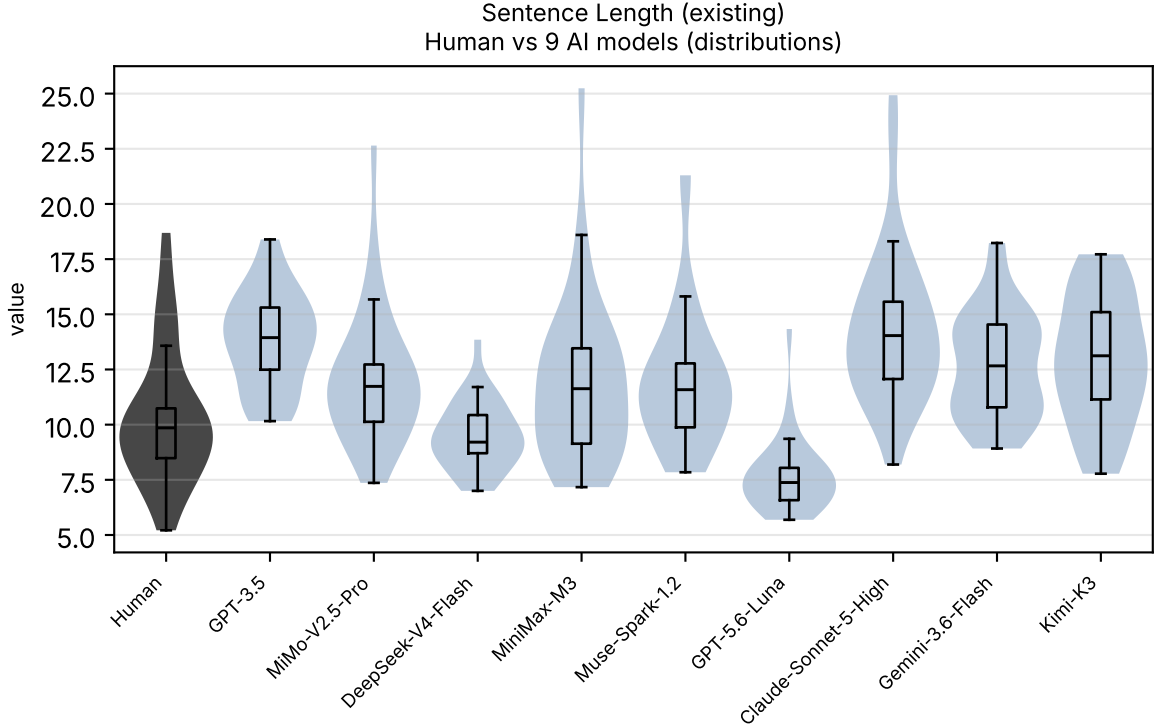

Within-group variance ratio (AI / Human)
< 1 = AI texts more uniform (* p<.05 t or Levene)
var ratio (log)
10^{0}
6×10^{-1}
4×10^{-1}
0.42*
0.78
0.47
1.18*
0.61
0.43
0.88*
0.75*
1.13
GPT-3.5
MiMo-V2.5-Pro
DeepSeek-V4-Flash
MiniMax-M3
Muse-Spark-1.2
GPT-5.6-Luna
Claude-Sonnet-5-High
Gemini-3.6-Flash
Kimi-K3

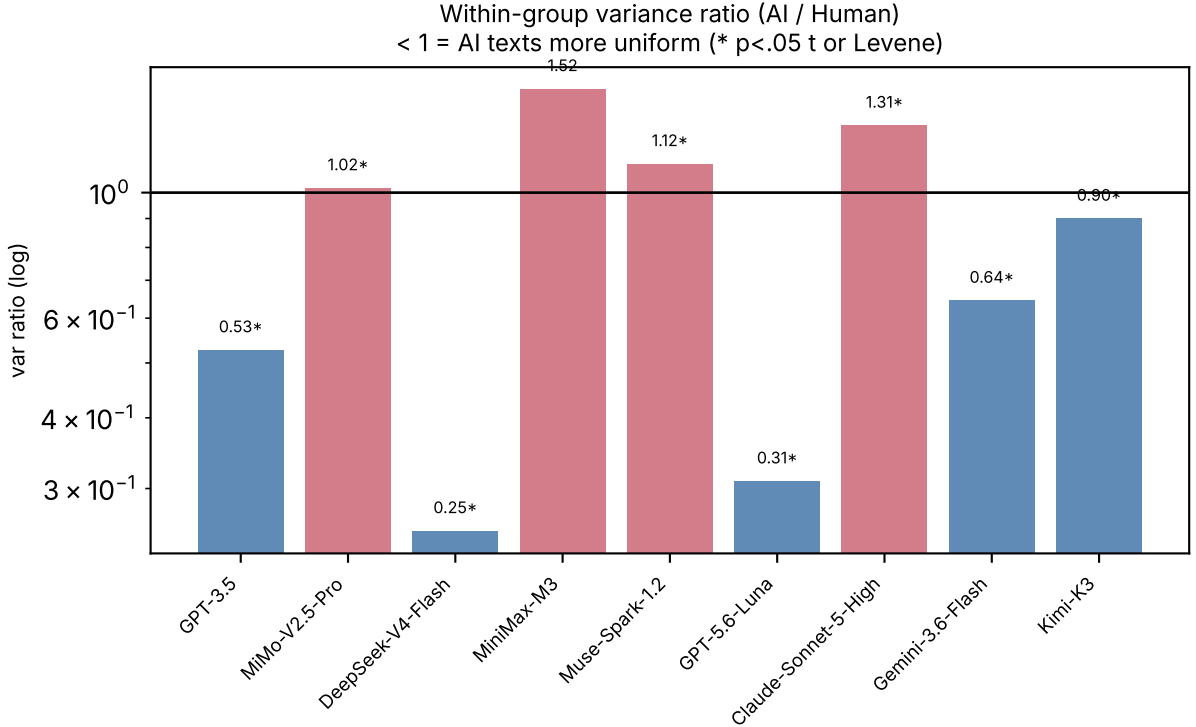

Sentences per Paragraph
Human vs 9 AI models (distributions)
value
60
50
40
30
20
10
0
Human
GPT-3.5
MiMo-V2.5-Pro
DeepSeek-V4-Flash
MiniMax-M3
Muse-Spark-1.2
GPT-5.6-Luna
Claude-Sonnet-5-High
Gemini-3.6-Flash
Kimi-K3

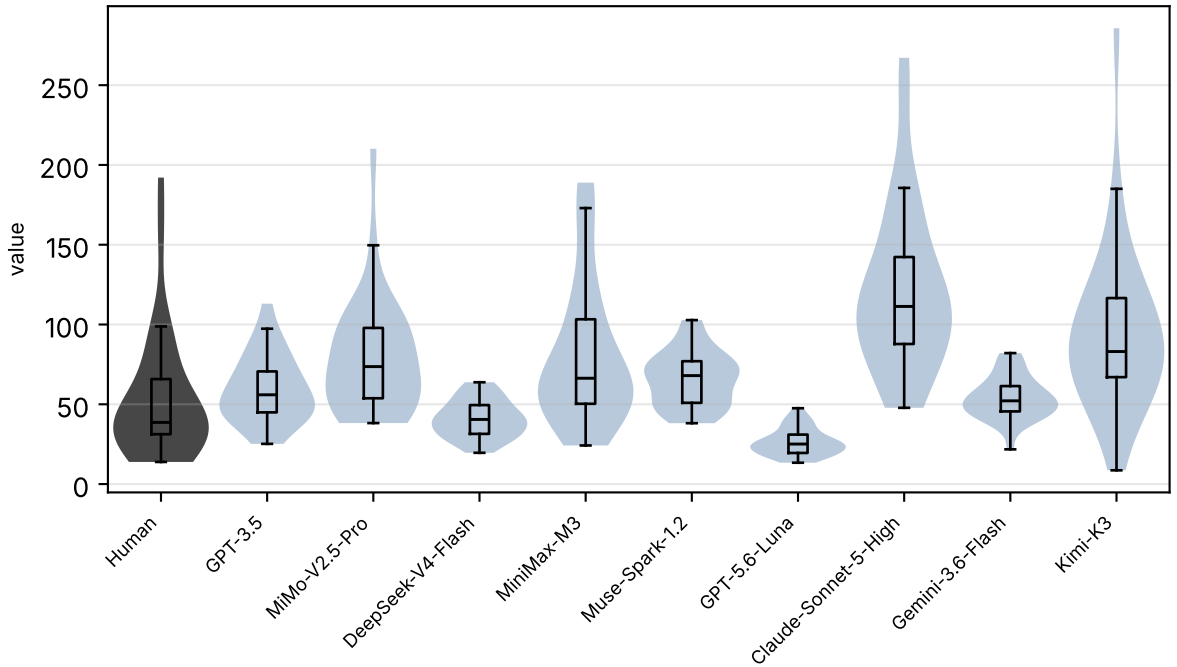

Within-group variance ratio (AI / Human)
< 1 = AI texts more uniform (* p<.05 t or Levene)
var ratio (log)
10^{0}
10^{-1}
10^{-2}
0.00
0.01
0.02
0.01
0.33
0.00*
0.00
0.00
0.01
GPT-3.5
MiMo-V2.5-Pro
DeepSeek-V4-Flash
MiniMax-M3
Muse-Spark-1.2
GPT-5.6-Luna
Claude-Sonnet-5-High
Gemini-3.6-Flash
Kimi-K3

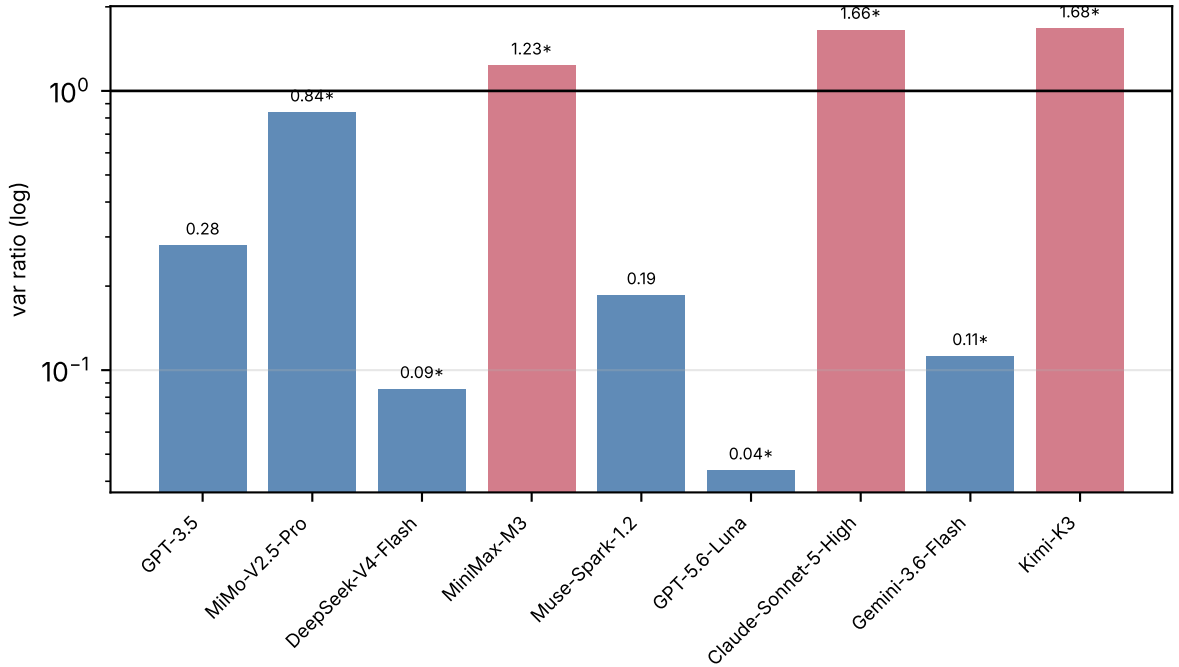

Word Length (existing)
Human vs 9 AI models (distributions)
value
5.5
5.0
4.5
4.0
3.5
Human
GPT-3.5
MiMo-V2.5-Pro
DeepSeek-V4-Flash
MiniMax-M3
Muse-Spark-1.2
GPT-5.6-Luna
Claude-Sonnet-5-High
Gemini-3.6-Flash
Kimi-K3

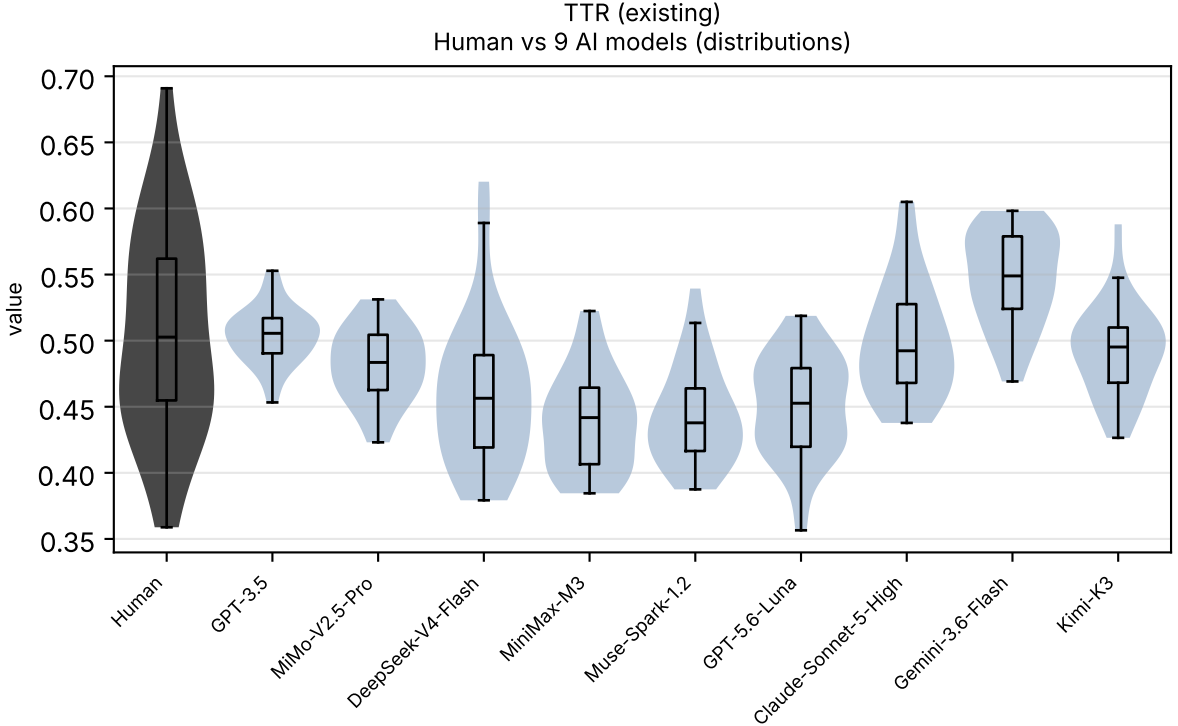

Within-group variance ratio (AI / Human)
< 1 = AI texts more uniform (* p<.05 t or Levene)
var ratio (log)
10^0
6×10^{-1}
0.51*
0.52
0.68
0.64
0.57
0.84*
1.13
0.51*
0.46*
GPT-3.5
MiMo-V2.5-Pro
DeepSeek-V4-Flash
MiniMax-M3
Muse-Spark-1.2
GPT-5.6-Luna
Claude-Sonnet-5-High
Gemini-3.6-Flash
Kimi-K3

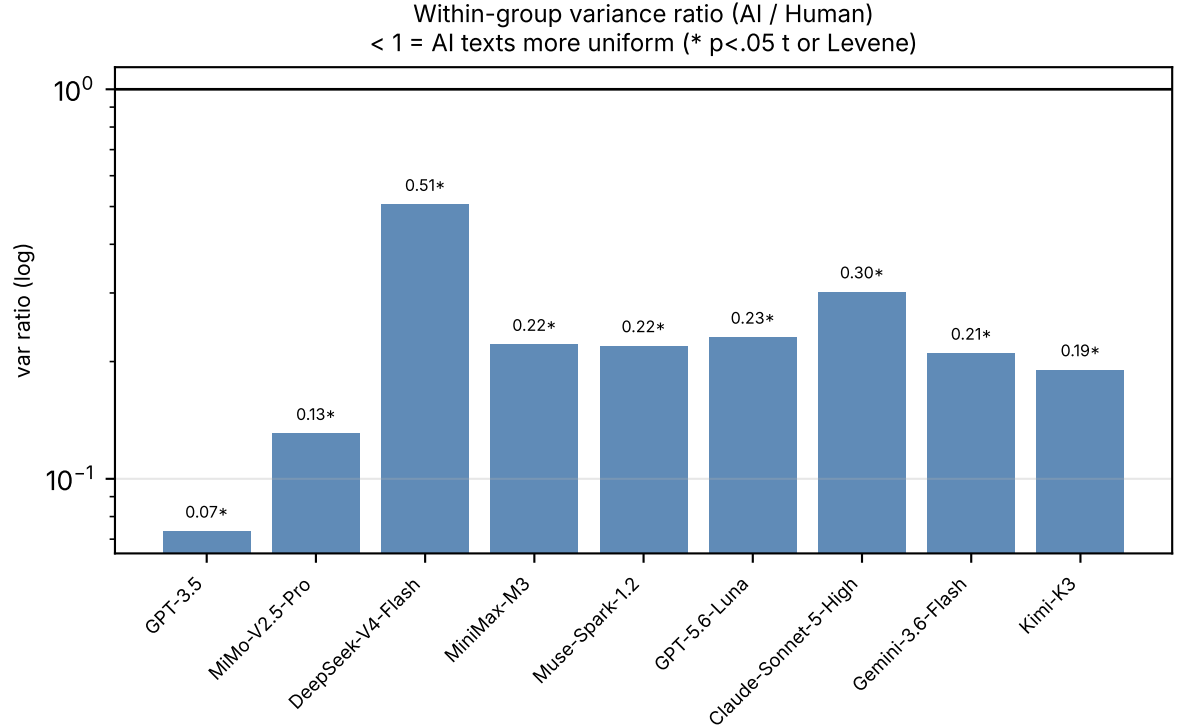

Sentence Length (existing)
Human vs 9 AI models (distributions)
value
25.0
22.5
20.0
17.5
15.0
12.5
10.0
7.5
5.0
Human
GPT-3.5
MiMo-V2.5-Pro
DeepSeek-V4-Flash
MiniMax-M3
Muse-Spark-1.2
GPT-5.6-Luna
Claude-Sonnet-5-High
Gemini-3.6-Flash
Kimi-K3

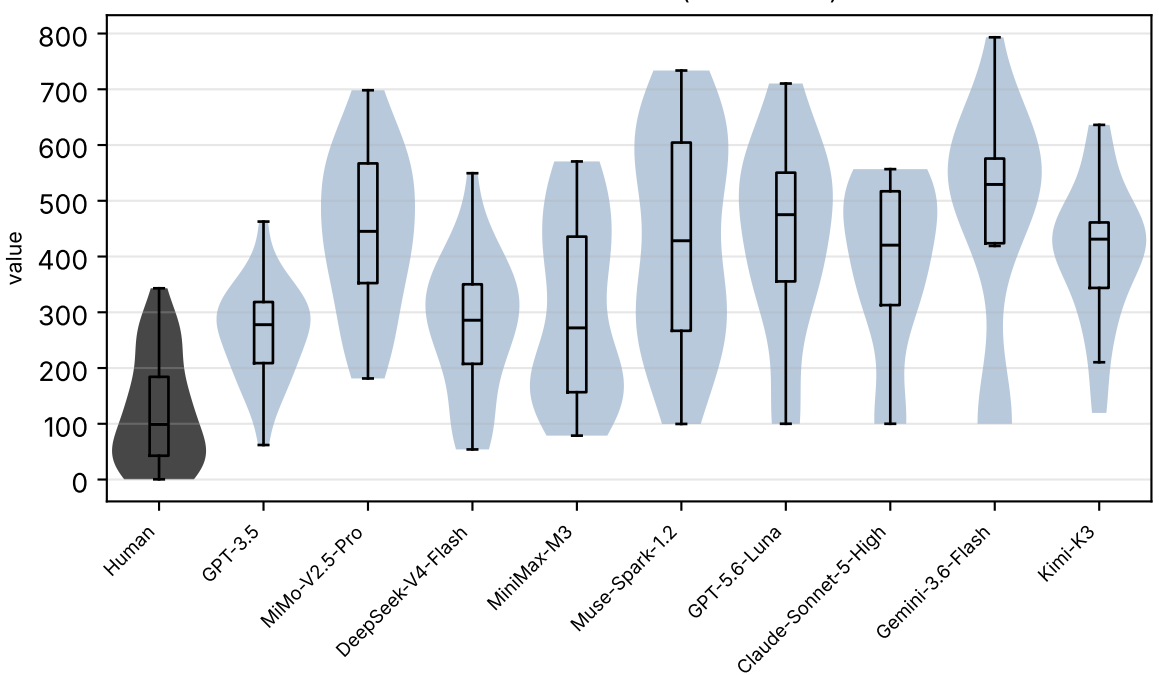

Within-group variance ratio (AI / Human)
< 1 = AI texts more uniform (* p<.05 t or Levene)
var ratio (log)
10^{0}
6×10^{-1}
4×10^{-1}
3×10^{-1}
0.53*
1.02*
0.25*
1.52
1.12*
0.31*
1.31*
0.64*
0.90*
GPT-3.5
MiMo-V2.5-Pro
DeepSeek-V4-Flash
MiniMax-M3
Muse-Spark-1.2
GPT-5.6-Luna
Claude-Sonnet-5-High
Gemini-3.6-Flash
Kimi-K3

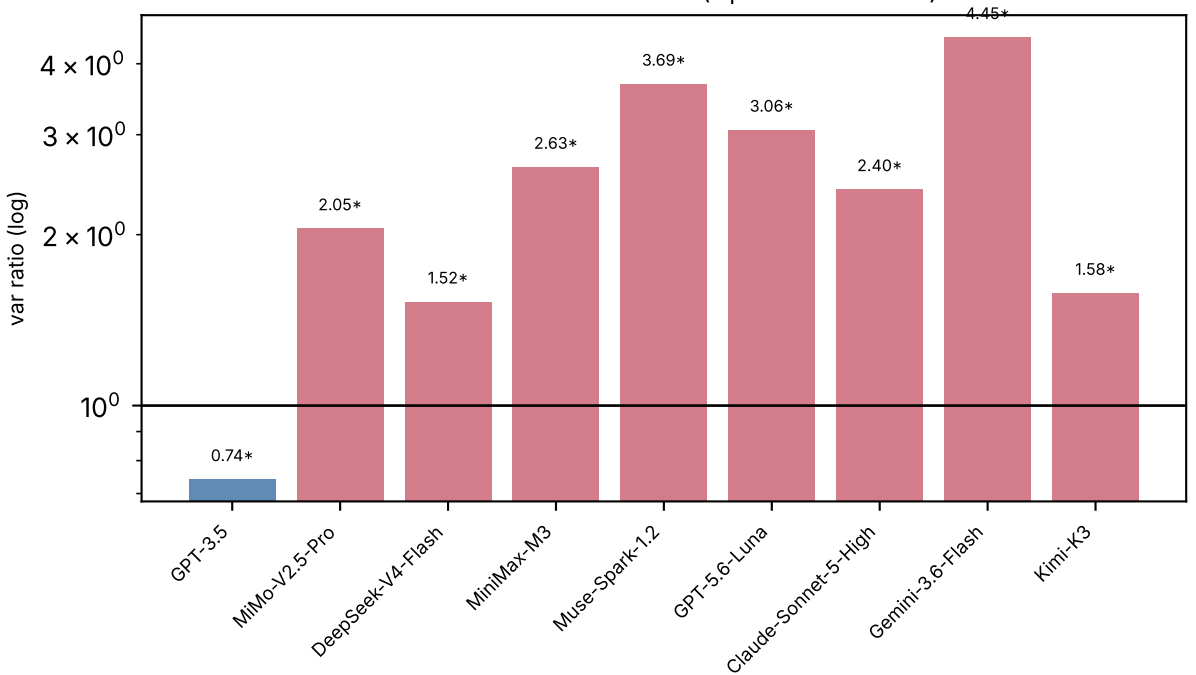

Sentence Length Variance (existing)
Human vs 9 AI models (distributions)
value
0
50
100
150
200
250
Human
GPT-3.5
MiMo-V2.5-Pro
DeepSeek-V4-Flash
MiniMax-M3
Muse-Spark-1.2
GPT-5.6-Luna
Claude-Sonnet-5-High
Gemini-3.6-Flash
Kimi-K3

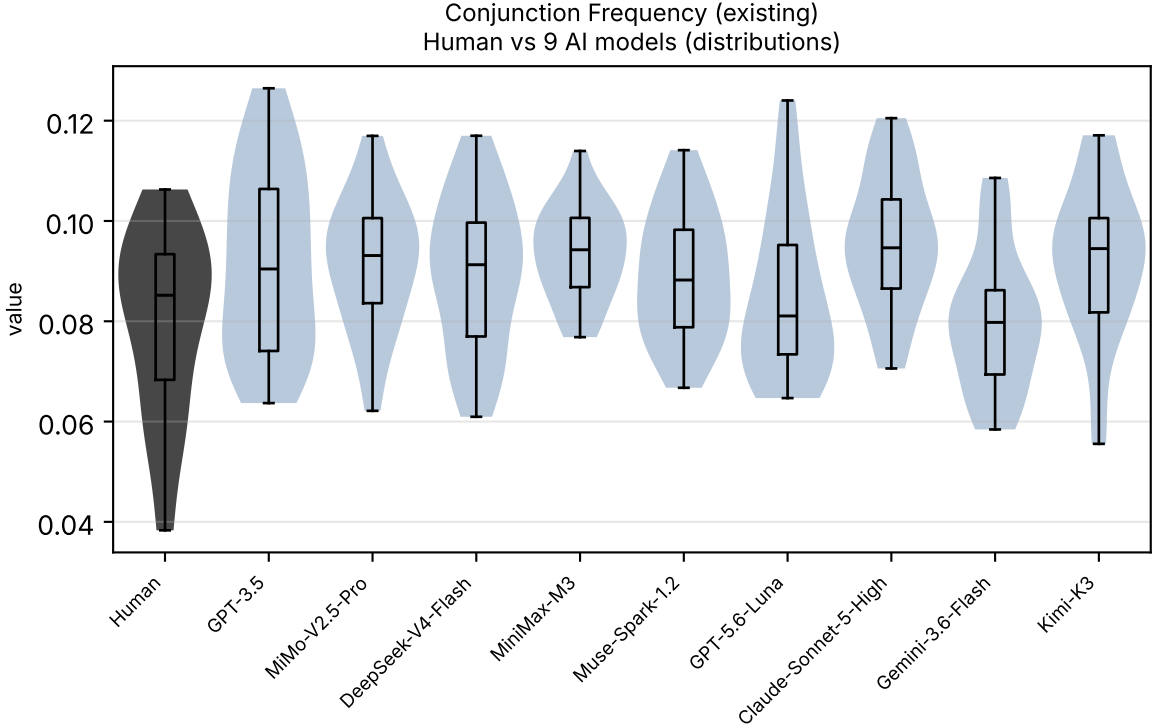

Within-group variance ratio (AI / Human)
< 1 = AI texts more uniform (* p<.05 t or Levene)
var ratio (log)
10^{0}
10^{-1}
0.28
0.84*
0.09*
1.23*
0.19
0.04*
1.66*
0.11*
1.68*
GPT-3.5
MiMo-V2.5-Pro
DeepSeek-V4-Flash
MiniMax-M3
Muse-Spark-1.2
GPT-5.6-Luna
Claude-Sonnet-5-High
Gemini-3.6-Flash
Kimi-K3

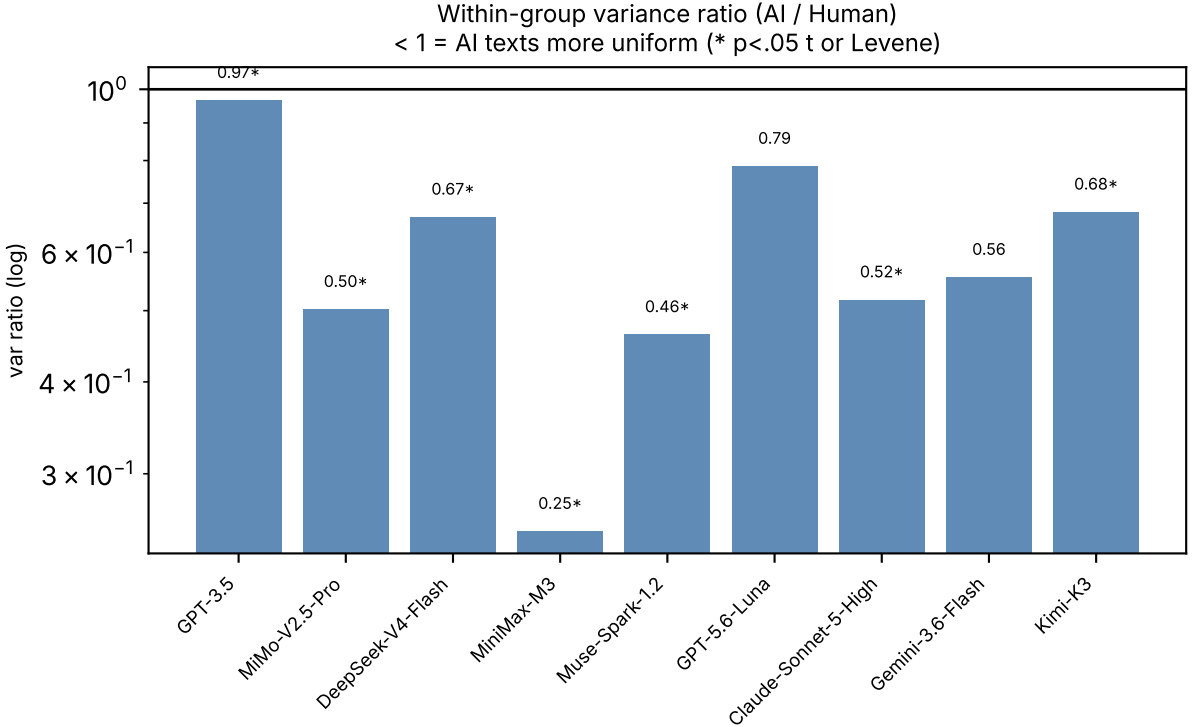

TTR (existing)
Human vs 9 AI models (distributions)
value
0.70
0.65
0.60
0.55
0.50
0.45
0.40
0.35
Human
GPT-3.5
MiMo-V2.5-Pro
DeepSeek-V4-Flash
MiniMax-M3
Muse-Spark-1.2
GPT-5.6-Luna
Claude-Sonnet-5-High
Gemini-3.6-Flash
Kimi-K3

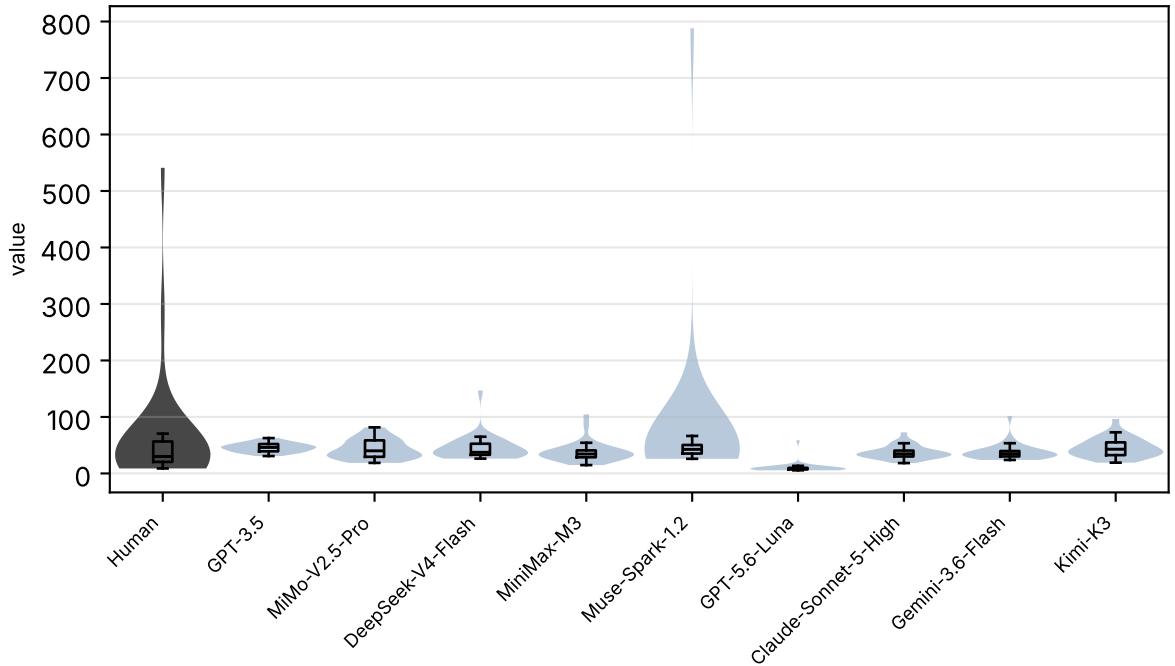

Within-group variance ratio (AI / Human)
< 1 = AI texts more uniform (* p<.05 t or Levene)
var ratio (log)
10^{0}
10^{-1}
0.07*
0.13*
0.51*
0.22*
0.22*
0.23*
0.30*
0.21*
0.19*
GPT-3.5
MiMo-V2.5-Pro
DeepSeek-V4-Flash
MiniMax-M3
Muse-Spark-1.2
GPT-5.6-Luna
Claude-Sonnet-5-High
Gemini-3.6-Flash
Kimi-K3

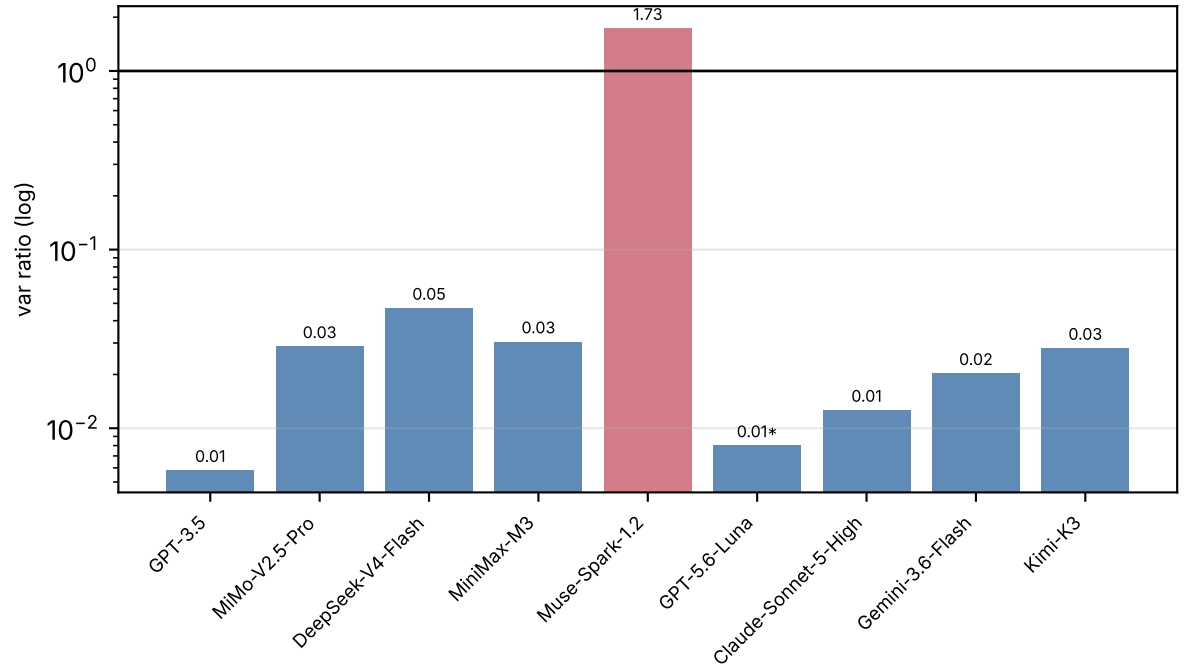

MTLD (existing)
Human vs 9 AI models (distributions)
value
800
700
600
500
400
300
200
100
0
Human
GPT-3.5
MiMo-V2.5-Pro
DeepSeek-V4-Flash
MiniMax-M3
Muse-Spark-1.2
GPT-5.6-Luna
Claude-Sonnet-5-High
Gemini-3.6-Flash
Kimi-K3

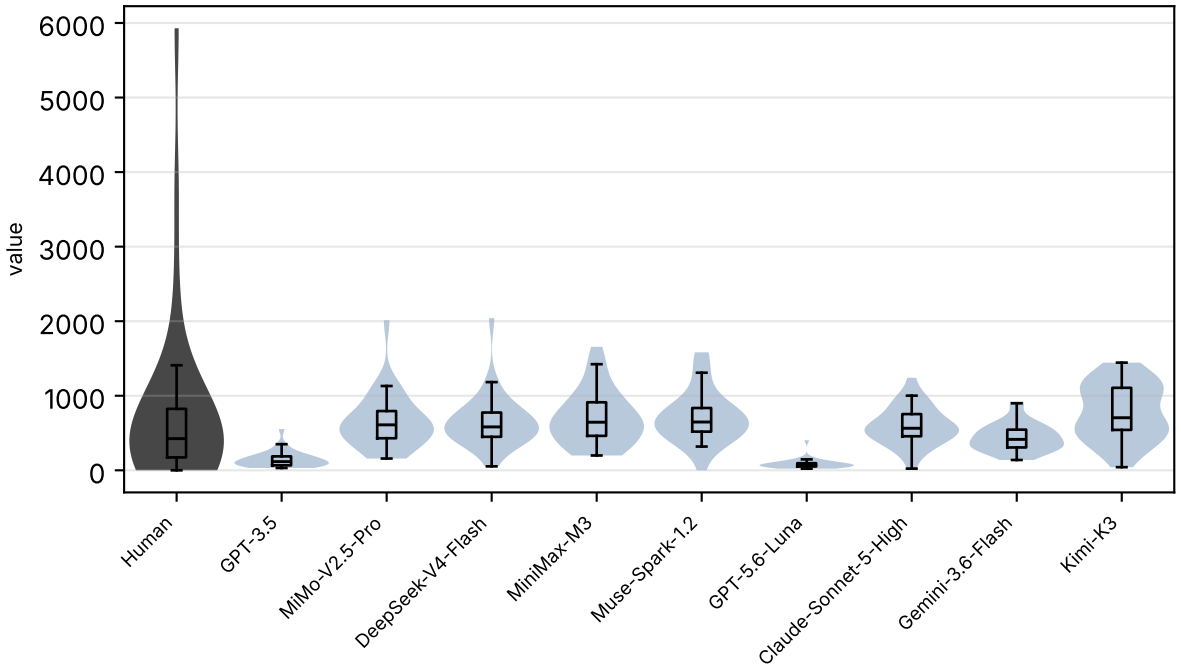

Within-group variance ratio (AI / Human)
< 1 = AI texts more uniform (* p<.05 t or Levene)
var ratio (log)
4 × 10^0
3 × 10^0
2 × 10^0
10^0
0.74*
2.05*
1.52*
2.63*
3.69*
3.06*
2.40*
4.45*
1.58*
GPT-3.5
MiMo-V2.5-Pro
DeepSeek-V4-Flash
MiniMax-M3
Muse-Spark-1.2
GPT-5.6-Luna
Claude-Sonnet-5-High
Gemini-3.6-Flash
Kimi-K3

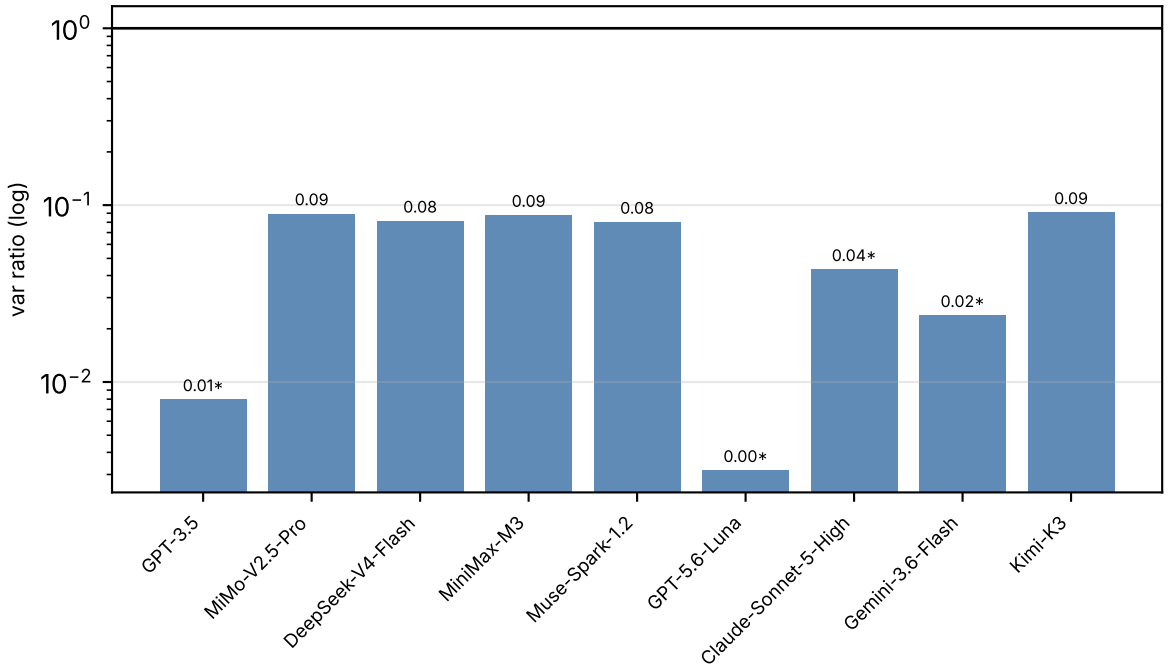

Conjunction Frequency (existing)
Human vs 9 AI models (distributions)
value
0.12
0.10
0.08
0.06
0.04
Human
GPT-3.5
MiMo-V2.5-Pro
DeepSeek-V4-Flash
MiniMax-M3
Muse-Spark-1.2
GPT-5.6-Luna
Claude-Sonnet-5-High
Gemini-3.6-Flash
Kimi-K3

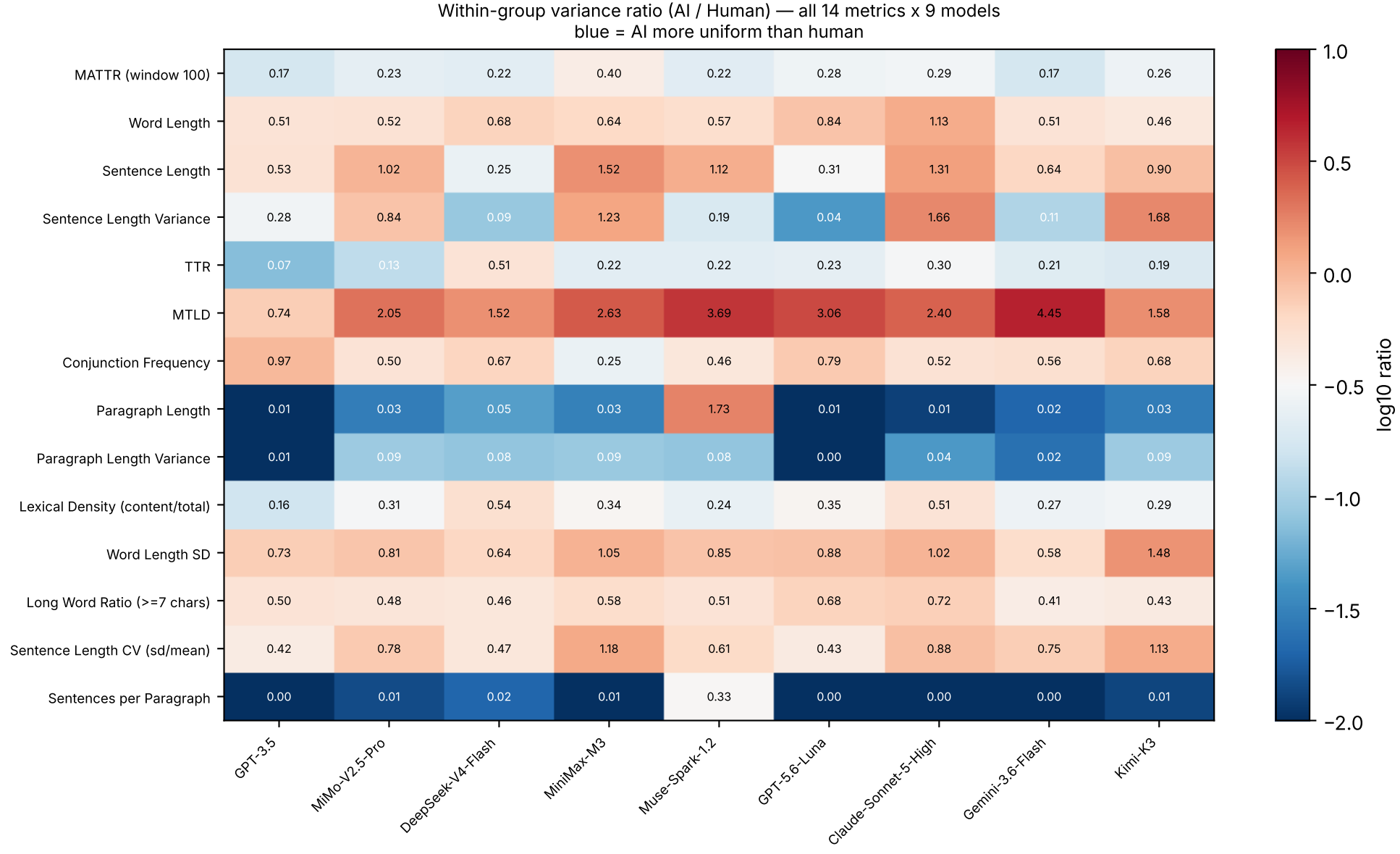

Within-group variance ratio (AI / Human)
< 1 = AI texts more uniform (* p<.05 t or Levene)
var ratio (log)
10^{0}
6×10^{-1}
4×10^{-1}
3×10^{-1}
0.97*
0.50*
0.67*
0.25*
0.46*
0.79
0.52*
0.56
0.68*
GPT-3.5
MiMo-V2.5-Pro
DeepSeek-V4-Flash
MiniMax-M3
Muse-Spark-1.2
GPT-5.6-Luna
Claude-Sonnet-5-High
Gemini-3.6-Flash
Kimi-K3

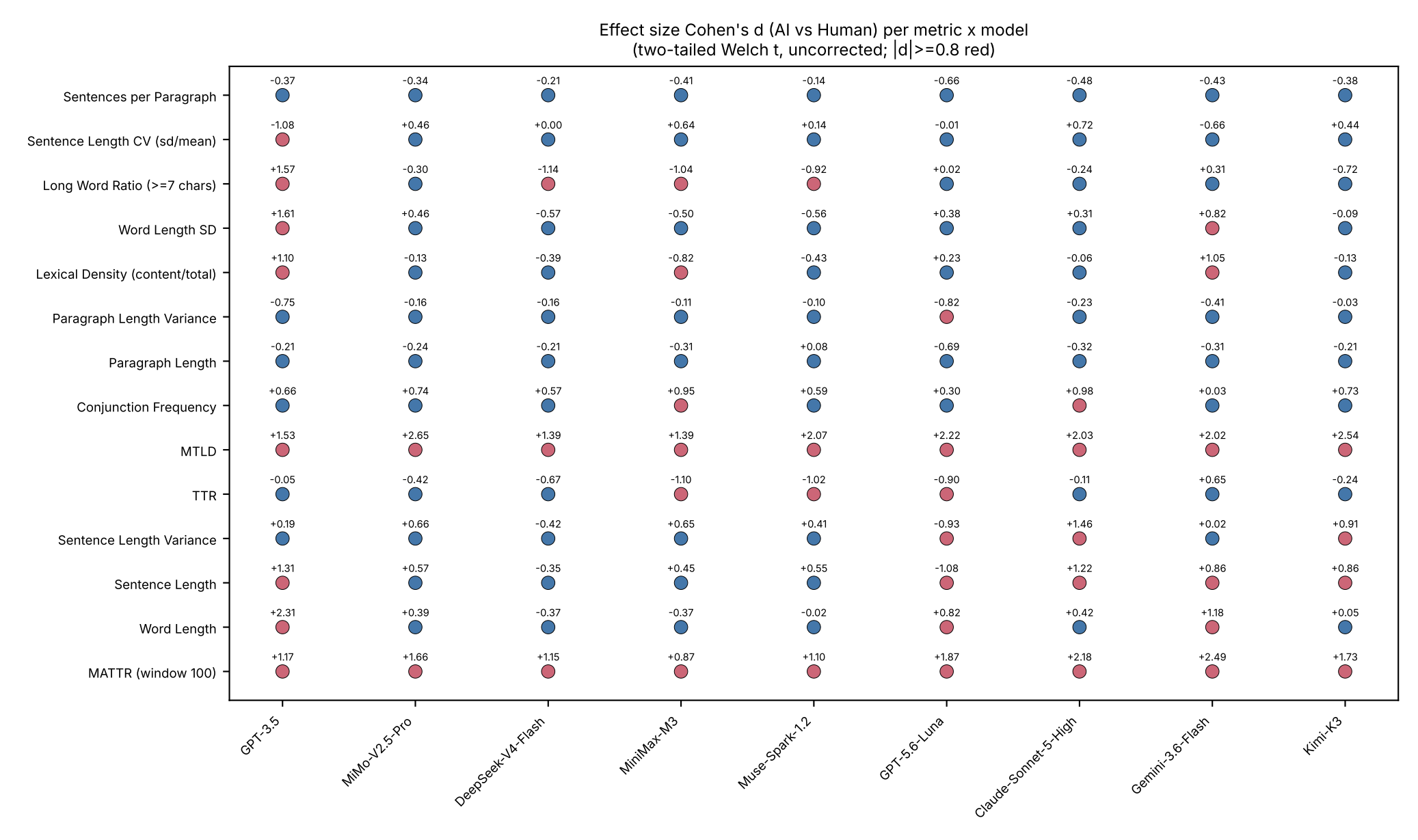

Paragraph Length (existing)
Human vs 9 AI models (distributions)
value
800
700
600
500
400
300
200
100
0
Human
GPT-3.5
MiMo-V2.5-Pro
DeepSeek-V4-Flash
MiniMax-M3
Muse-Spark-1.2
GPT-5.6-Luna
Claude-Sonnet-5-High
Gemini-3.6-Flash
Kimi-K3


Within-group variance ratio (AI / Human)
< 1 = AI texts more uniform (* p<.05 t or Levene)
var ratio (log)
10^{0}
10^{-1}
10^{-2}
1.73
0.01
0.03
0.05
0.03
0.01*
0.01
0.02
0.03
GPT-3.5
MiMo-V2.5-Pro
DeepSeek-V4-Flash
MiniMax-M3
Muse-Spark-1.2
GPT-5.6-Luna
Claude-Sonnet-5-High
Gemini-3.6-Flash
Kimi-K3


Paragraph Length Variance (existing)
Human vs 9 AI models (distributions)
value
6000
5000
4000
3000
2000
1000
0
Human
GPT-3.5
MiMo-V2.5-Pro
DeepSeek-V4-Flash
MiniMax-M3
Muse-Spark-1.2
GPT-5.6-Luna
Claude-Sonnet-5-High
Gemini-3.6-Flash
Kimi-K3


Within-group variance ratio (AI / Human)
< 1 = AI texts more uniform (* p<.05 t or Levene)
var ratio (log)
10^{0}
10^{-1}
10^{-2}
0.01*
0.09
0.08
0.09
0.08
0.00*
0.04*
0.02*
0.09
GPT-3.5
MiMo-V2.5-Pro
DeepSeek-V4-Flash
MiniMax-M3
Muse-Spark-1.2
GPT-5.6-Luna
Claude-Sonnet-5-High
Gemini-3.6-Flash
Kimi-K3


Within-group variance ratio (AI / Human) — all 14 metrics x 9 models
blue = AI more uniform than human
MATTR (window 100) 0.17 0.23 0.22 0.40 0.22 0.28 0.29 0.17 0.26
Word Length 0.51 0.52 0.68 0.64 0.57 0.84 1.13 0.51 0.46
Sentence Length 0.53 1.02 0.25 1.52 1.12 0.31 1.31 0.64 0.90
Sentence Length Variance 0.28 0.84 0.09 1.23 0.19 0.04 1.66 0.11 1.68
TTR 0.07 0.13 0.51 0.22 0.22 0.23 0.30 0.21 0.19
MTLD 0.74 2.05 1.52 2.63 3.69 3.06 2.40 4.45 1.58
Conjunction Frequency 0.97 0.50 0.67 0.25 0.46 0.79 0.52 0.56 0.68
Paragraph Length 0.01 0.03 0.05 0.03 1.73 0.01 0.01 0.02 0.03
Paragraph Length Variance 0.01 0.09 0.08 0.09 0.08 0.00 0.04 0.02 0.09
Lexical Density (content/total) 0.16 0.31 0.54 0.34 0.24 0.35 0.51 0.27 0.29
Word Length SD 0.73 0.81 0.64 1.05 0.85 0.88 1.02 0.58 1.48
Long Word Ratio (>=7 chars) 0.50 0.48 0.46 0.58 0.51 0.68 0.72 0.41 0.43
Sentence Length CV (sd/mean) 0.42 0.78 0.47 1.18 0.61 0.43 0.88 0.75 1.13
Sentences per Paragraph 0.00 0.01 0.02 0.01 0.33 0.00 0.00 0.00 0.01
GPT-3.5
MiMo-V2.5-Pro
DeepSeek-V4-Flash
MiniMax-M3
Muse-Spark-1.2
GPT-5.6-Luna
Claude-Sonnet-5-High
Gemini-3.6-Flash
Kimi-K3
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
log10 ratio


Effect size Cohen's d (AI vs Human) per metric x model
(two-tailed Welch t, uncorrected; |d|>=0.8 red)
Sentences per Paragraph
-0.37 -0.34 -0.21 -0.41 -0.14 -0.66 -0.48 -0.43 -0.38
Sentence Length CV (sd/mean)
-1.08 +0.46 +0.00 +0.64 +0.14 -0.01 +0.72 -0.66 +0.44
Long Word Ratio (>=7 chars)
+1.57 -0.30 -1.14 -1.04 -0.92 +0.02 -0.24 +0.31 -0.72
Word Length SD
+1.61 +0.46 -0.57 -0.50 -0.56 +0.38 +0.31 +0.82 -0.09
Lexical Density (content/total)
+1.10 -0.13 -0.39 -0.82 -0.43 +0.23 -0.06 +1.05 -0.13
Paragraph Length Variance
-0.75 -0.16 -0.16 -0.11 -0.10 -0.82 -0.23 -0.41 -0.03
Paragraph Length
-0.21 -0.24 -0.21 -0.31 +0.08 -0.69 -0.32 -0.31 -0.21
Conjunction Frequency
+0.66 +0.74 +0.57 +0.95 +0.59 +0.30 +0.98 +0.03 +0.73
MTLD
+1.53 +2.65 +1.39 +1.39 +2.07 +2.22 +2.03 +2.02 +2.54
TTR
-0.05 -0.42 -0.67 -1.10 -1.02 -0.90 -0.11 +0.65 -0.24
Sentence Length Variance
+0.19 +0.66 -0.42 +0.65 +0.41 -0.93 +1.46 +0.02 +0.91
Sentence Length
+1.31 +0.57 -0.35 +0.45 +0.55 -1.08 +1.22 +0.86 +0.86
Word Length
+2.31 +0.39 -0.37 -0.37 -0.02 +0.82 +0.42 +1.18 +0.05
MATTR (window 100)
+1.17 +1.66 +1.15 +0.87 +1.10 +1.87 +2.18 +2.49 +1.73
GPT-3.5
MiMo-V2.5-Pro
DeepSeek-V4-Flash
MiniMax-M3
Muse-Spark-1.2
GPT-5.6-Luna
Claude-Sonnet-5-High
Gemini-3.6-Flash
Kimi-K3